Final Report for e-Portfolio including reflective

(*Research Methods and Professional Practice*)

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e-Portfolio Digital Report

Research Methods and Professional Practice module digital e-portfolio (hosted at: https://mariamalmarzooqiessex.github.io/portfolio/)

Reflective Commentary: Research Methods and Professional Practice

The purpose of the Research Methods and Professional Practice module which I undertook was to crystallize my learning processes. Each module discussed individual components of the research process, and complimented it with practical activities, and managerial frameworks of ethics in research, and professional development. Working through these components facilitated the development of my analytical and professional skills. Below are reflections from the modules, and these are directed to my e-portfolio to justify these learnings.

Unit 1: Research Methods and Ethics in Computing

This module examined what techniques and frameworks are used to conduct research. The module examined the difference between inductive and deductive reasoning. \As reasoning is a fundamental question for all research projects, it is important to be able to process all the different components of reasoning (Babbie, 2020). One of these components is ethics, and I used reflective activities to explore the ethics of new technologies. I focused my reflective activities on the ethics of new technologies, in particular new digital technologies, and on the components of privacy, bias, and accountability.

Unit 2: Research Questions, Literature Review, and Proposal

During this unit, I learned to reshape broad research interests into constructive research questions. For example, when analyzing a particular case study, I understood the significance of constructing a relevant example. Constructive peer feedback helped refine the draft questions and the proposal templates, as well as the literature review outline, providing pathways to think coherently, while the critical review of past research studies helped identify research gaps and question assumptions.

Unit 3: Methodology and Research Methods

This unit demonstrated how research objectives correlate with specific methodologies and how to structure data collection methods. I assessed the three methodologies—qualitative, quantitative, and mixed—and studied research philosophies, including positivism and interpretivism (Bryman, 2016). Appreciating the significance of philosophical congruence underscored the role of the difference that a researcher’s methodological choice made to the research outcomes. The review of a range of research designs boosted my confidence in making certain methodological choices, and I look forward to putting these principles to use, both in my studies and practice.

Unit 4: Case Studies, Focus Groups, and Observations

In this unit, I studied qualitative research methodologies, particularly focusing on case studies, focus group discussions, and the various approaches of conducting observational research. I analyzed the various instances in which each method would apply and how each contributes to a deeper understanding of user behaviour (Creswell and Creswell, 2018). Engaging with qualitative techniques enabled me to document attitudes and experiences, which, even though primarily of interest to me, I found valuable. I found focus groups particularly useful, and I plan to use this method in future research that examines the diverse perspectives of stakeholders in UX and HCI.

Unit 5: Interviews, Surveys, and Questionnaire Design

This unit taught me about the intricacies involved in designing effective data-collection instruments. I learned the importance of survey design and structure, sampling, and clarity to avoid leading questions, and even to the point of causing ethical dilemmas. Real-world case examples helped to remind me of the ethical responsibilities of research. I have been able to survey myself and apply the principles of the design to some extent, which has really helped me focus on the design of the survey, the structure, the questions, and the overall neutrality, which in turn improved the quality of my research and writing.

Unit 6: Quantitative Methods – Descriptive and Inferential Statistics

I feel far more confident with numerical data and the techniques involved. I used descriptive measures such as the mean, median, standard deviation, and range to help me understand the datasets (Saunders, Lewis, and Thornhill, 2019). It improved my analytical thinking to consider what values indicated about other values and what trends they represented. I was able to further develop my data-related skills and support evidence-based decisions after using Excel and R multiple times.

Unit 7: Inferential Statistics and Hypothesis Testing

In this section, I acquired my first set of skills in inferential statistics, confidence intervals, probability distributions, and hypothesis testing. I had to run a t-test, and from there I was able to proceed with p-values and related concepts regarding statistically significant results. This was my first exposure to analytical risk, and I had to learn about Type I and Type II errors. I feel this has given me the knowledge to handle more complex future assignments, such as experimental analyses and machine learning assessments.

Unit 8: Data Analysis and Visualisation

I learnt to display data in a way that is suitable for this unit. I studied a number of visualisation methods and learned about the suitability of certain chart types for given data and the objectives of a particular study. I was able to improve my understanding of data and analytical insights in a way that is understandable to a range of people, from non-experts to more sophisticated audiences, thanks to my knowledge of how visual displays can affect how the data is understood (Walliman, 2018).

Over this unit, I began seeing data visuals as narrative tools rather than mere displays. I now see charts as more than mere illustrations; they are guides for shaping readers' attention to specific interpretations of data. Shaping dashboards and summary visuals became particularly important for communicating insights to audiences with little or no technical background. I began asking myself whether a visual aid detracts from the narrative's focus.

Unit 9: Validity and Generalisability in Research

This unit improved my ability to assess research quality. I examined in some detail the concepts of internal and external validity, reliability, and generalisability, and their impact on a research study's credibility (Correa, Simmons, and Taylor, 2023). I also looked back at the roles of sample size, research design, and accurate measurement in producing trustworthy findings. I applied these concepts to my literature review and proposal, which improved my source selection and the design of my experiment, thereby again highlighting the importance of precision and the ability to replicate.

Unit 10: Research Writing

This unit offered more support for research writing. I learned how to build a more structured argument, maintain a more logical flow, and better integrate paraphrasing and quoting into my explanations. The feedback on my proposal highlighted and improved clarity, flow, and consistency. The practice I did with the Harvard style of referencing contributed to my improvement in formal writing and gave me some preparation for the writing I will need to in my dissertation, proposals for funding, and in professional reports.

Unit 11: Going Forward – Professional Development and e-Portfolio

This unit concentrated on the processes of reflection and planning for the future. Preparing a Personal Development Plan and a Skills Matrix prompted me to recognize my achievements in time management, collaboration, and ethical awareness (Deckard, 2023). Building my e-portfolio enabled me to articulate my academic identity and provided me with a valuable tool for employment and networking opportunities.

Unit 12: Project Management and Risk

In the final unit, I focused on project planning tools, including Gantt charts, risk matrices, and collaborative tools. I learned to differentiate the Agile and Waterfall methodologies and the importance of early project risk identification. This knowledge will help me during my MSc capstone project and in future project-based technical roles.

Conclusion

The Research Methods and Professional Practice module has been a primary cornerstone of my whole academic experience so far. It provided me with a guide to complete every step of the research process, from developing a set of constructive and focused research questions to applying research methodologies that are both ethically sound and methodologically defensible. With each step of the module, I improved my technical skills and my ability to perform critical evaluations.

One of the biggest strengths of the module was its consistently strong emphasis on the research's ethical, legal, and socially responsible aspects. Working on the research proposal addressing the contemporary issues of the unethical and irresponsible use of data and generative AI, I reinforced transparency, accountability, and responsible practices in computing. All of these will be more relevant and will strengthen further as I will be dealing with complicated and sensitive systems in the future.

The e-portfolio functions as both a learning archive and a professional showcase. It also fosters structured reflection and the articulation of research concepts, as well as the e-portfolio and, more recently, the e-portfolio. I intend to apply the skills of data analysis, hypothesis testing, and project planning to my MSc capstone project and other optional research roles. This module has enabled me to be an ethically and socially confident computing professional.

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Reflective Report

Learning Journey and Activities (WHAT)

Prior to starting this module, my understanding of research was largely theoretical, confined to a few basic academic principles. I had almost no exposure to designing measurable research projects or applying any of the formal research methods. I had only a very surface-level understanding of the development of research questions, the process of data analysis, or the assessment of findings. This module paved the way for a comprehensive understanding of research as a field of study and as a practice (Anderson and Hepburn, 2020).

Unit 1 outlined some of the basic building blocks of the Scientific Method. These included both the inductive and deductive approaches to reasoning, which enhanced my ability to process information both analytically and logically. This unit also drew attention to the lack of ethics within research, particularly within the field of computing, and stressed the importance of a research practitioner's responsibility, ethical choices, and the enforcement of a code of ethics (Boza, 2022).

In Unit 2, the emphasis was on developing research questions and planning research proposals. I initially struggled to narrow a broad field of interest to a point where I could formulate a few concise, researchable questions (LaMorte, 2018). I learned to convert abstract concepts into measurable objectives through guided activities and discussions with my colleagues. After being exposed to techniques for literature reviews, I was able to organize my academic resources to better articulate research gaps.

In units 3 to 5, I studied different research philosophies and methodologies, broadening my understanding of the influence of various ontological and epistemological perspectives on research design (Saunders, Lewis, and Thornhill, 2012). I also gained first-hand experience with qualitative data collection methods, including case studies, interviews, and focus groups, and learned to assess and appreciate the advantages and disadvantages of each.

In units 6 to 8, I learned about quantitative analysis and acquired skills in descriptive and inferential statistics, including confidence intervals and hypothesis testing. I was also introduced to some of the principles of data visualisation, which enhanced my ability to present research findings. Unit 9 taught me the importance of validity, reliability, and generalisation in producing credible and replicable research.

The final units emphasised research writing, professional development, and project management. Creating an e-Portfolio encouraged reflective practice (Microsoft, 2023), while learning about project risk and change control enhanced my strategic planning and execution skills.

Challenges, Emotions, and Growth (SO WHAT)

This module, like the previous one, has brought me new academic and emotional hurdles, allowing me to develop personally and professionally. Understanding the different philosophies of research and their corresponding methodologies, including the stats associated with each, was one of the toughest tasks. This, like many other things, was the first time I had encountered something like this. As I have said before, formulating research questions, interpreting and drawing meaning from, or defining quantifiable data, traverses a field of uncertainty. These uncertainties demonstrate the lack of knowledge I hold, and illustrate, in this case, a more effective line of thinking and, in this instance, the absence of organized, controlled learning or the need for greater discipline.

The constructive and informed synthesis of research over time, along with the positive feedback, the performed empirical tests (for me, stats), and the guided research synthesis toward achieving the defined metrics improved confidence, and more generally, the ability to intellectually and emotionally transform the constructive state of confusion to clarity. The positive reinforcement of this transition state helped me emotionally and intellectually. Acknowledging the importance of time critically, without thinking of it as a constraint, has helped me manage the numerous, and, in the end, several are astutely the same, within an ordered structure of research synthesis (LaMorte, 2018).

This made me feel positive, and from an emotional and academic standpoint, personally, the most influential in overcoming the hurdles. Throughout this experience, I have reflected more on the issue and improved my problem-solving from a research perspective and in a reflective, autonomous-learning state.

Skills Gained and Future Applications (NOW WHAT)

- Devise focused, ethical, and clear research question(s)
- Develop research methodologies congruent with the different philosophical views
- Assess and review the literature to determine and articulate the gaps
- Defend the selection of relevant data collection methods

- Employ inferential and descriptive statistical techniques
- Interpret and examine data and communicate results with suitable visual representations
- Prepare coherent and academically rigorous research papers

My soft skill targets have also improved:

- Critical thinking: Analyzing and questioning evidence and sources
- Communication: Articulating and documenting research efficiently and clearly in written and oral form
- Time management: Maintaining regular updates on the e-Portfolio and efficiently prioritizing tasks
- Ethical awareness: Applying research responsibilities, risks, and rights

Professional Development Plan (PDP)

Goal	**Action Steps**	**Action Steps**	**Timeline**	**Success Criteria**
Improve qualitative data analysis skills	Complete training in NVivo and thematic analysis	NVivo software, online courses (e.g., FutureLearn)	1 month	Ability to code and analyse interview transcripts
Strengthen academic writing for dissertation	Review past work and seek supervisor feedback	Academic journals, Grammarly, tutor support	Ongoing	Clear, structured, and coherent academic drafts
Build project management knowledge	Enrol in PRINCE2 or Agile foundation course	Course materials and study time	3 month	Successful completion of certification

Skills Matrix

Skill Area	Before Module	After Module	Evidence
Research Design	Limited	Competent	Developed proposal draft and completed peer feedback
Ethics in Computing	Unfamiliar	Aware and confident	Menlo-based ethics document for simulated research project
Data Collection Methods	Basic	Informed and strategic	Case study and interview planning exercises
Quantitative Analysis	Insecure	Proficient	Statistical hypothesis testing using sample datasets
Data Visualisation	Novice	Practical	Designed bar charts and analysed dashboard outputs
Academic Writing	Developing	Clear and structured	Final research proposal submission
Reflective Thinking	Superficial	Deep and purposeful	Weekly e-Portfolio reflections and peer interaction
Time and Task Management	Reactive	Organised	Unit-by-unit progress tracking and milestone review

Assessment of Learning Outcomes and Evidence

Each unit contributed to developing a broader and more advanced understanding of research.

- **Units 1–3:** Established ethical awareness and research philosophy foundations.
- **Units 4–5:** Developed technical competence in data collection design.
- **Units 6–8:** Improved statistical understanding and data visualisation skills.
- **Units 9–10:** Enhanced accuracy, validation, and reporting capabilities.
- **Units 11–12:** Encouraged professional self-reflection and risk management.

Annotated readings, proposals, and ethics documentation serve as evidence of applied learning.

Conclusion

This module has definitely been one of the most beneficial academic developments I have had the opportunity of experiencing. It evolved beyond simply cultivating my research skills to encompass reflection, ethics, and professional development. The fusion of hard and soft skills has primed me for my MSc project and the academic and professional hurdles that lie ahead. Having acquired the skills, I am ready to embark on a journey of independent research and to continue my evolution as a reflective practitioner.

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